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TINKERCAD

LED

```
// C++ code

//

void setup()

{

  pinMode(13, OUTPUT);

}

void loop()

{

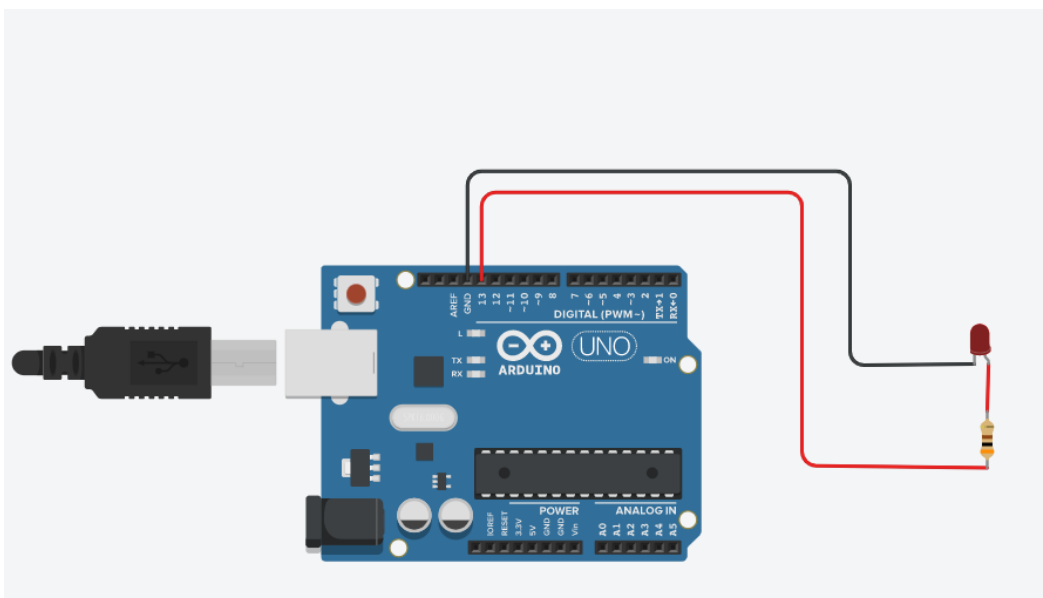
  digitalWrite(13, HIGH);

  //delay(1000); // Wait for 1000 millisecond(s)

  ////digitalWrite(LED_BUILTIN, LOW);

  //delay(1000); // Wait for 1000 millisecond(s)

}
```



SEMÁFORO

```
int vermelho = 10;

int amarelo = 9;

int verde = 8;


void setup() {

    // indicando para o arduíno quais portas vamos usar

    pinMode(vermelho, OUTPUT);

    pinMode(amarelo, OUTPUT);

    pinMode(verde, OUTPUT);

}


void loop()

{

    digitalWrite(vermelho, LOW);

    digitalWrite(amarelo, HIGH);

    digitalWrite(verde, LOW);


    delay(2000);


    digitalWrite(amarelo, LOW);

    digitalWrite(vermelho, HIGH);

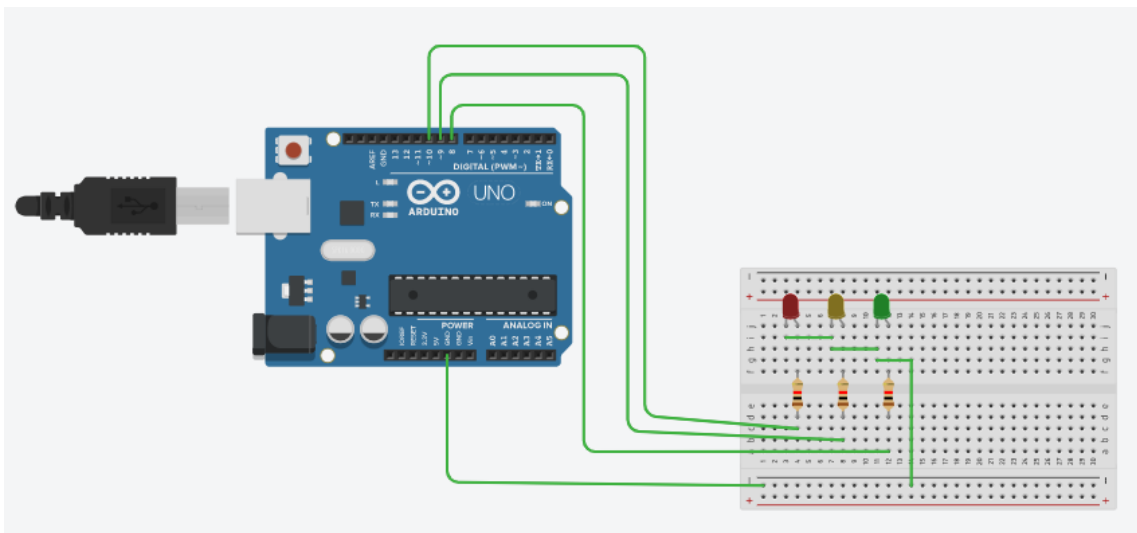

    // esperamos 5s com o sinal fechado

    delay(5000);
```

```
digitalWrite(verde, HIGH);  
  
digitalWrite(amarelo, LOW);  
  
digitalWrite(vermelho, LOW);
```

```
delay(5000);
```

```
}
```



LED RGB

```
// Piscando o LED RGB

void setup() {

  pinMode(11, OUTPUT); // vermelho

  pinMode(12, OUTPUT); //verde

  pinMode(13, OUTPUT); //azul

}

void loop() {

  delay(50);

  digitalWrite(11, HIGH);

  delay(50);

  digitalWrite(11, LOW);

  delay(50);

  digitalWrite(12, HIGH);

  delay(50);

  digitalWrite(12, LOW);

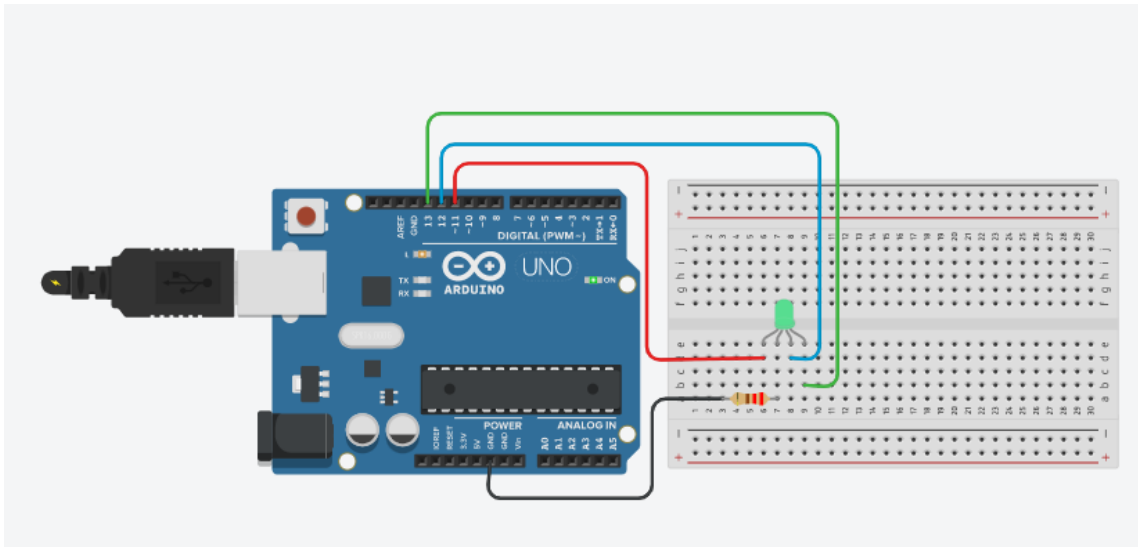
  delay(50);

  digitalWrite(13, HIGH);

  delay(50);

  digitalWrite(13, LOW);

}
```



LCD

```
#include <LiquidCrystal.h> //Biblioteca do Cristal Líquido
```

```
LiquidCrystal lcd(2,4,10,11,12,13); //Criação do objeto lcd com os pinos que conectam os displays
```

```
void setup()
```

```
{
```

```
  lcd.begin(16,2); //Inicia a comunicação do display com quantidade de colunas e linhas
```

```
  lcd.print("Eder!"); //Imprime o nome Eder no Display
```

```
  //lcd.clear(); //Limpa a tela colocando o cursor na posição (0,0)
```

```
  lcd.setCursor(0,1); //Coloca o cursor na coluna e linha que quer
```

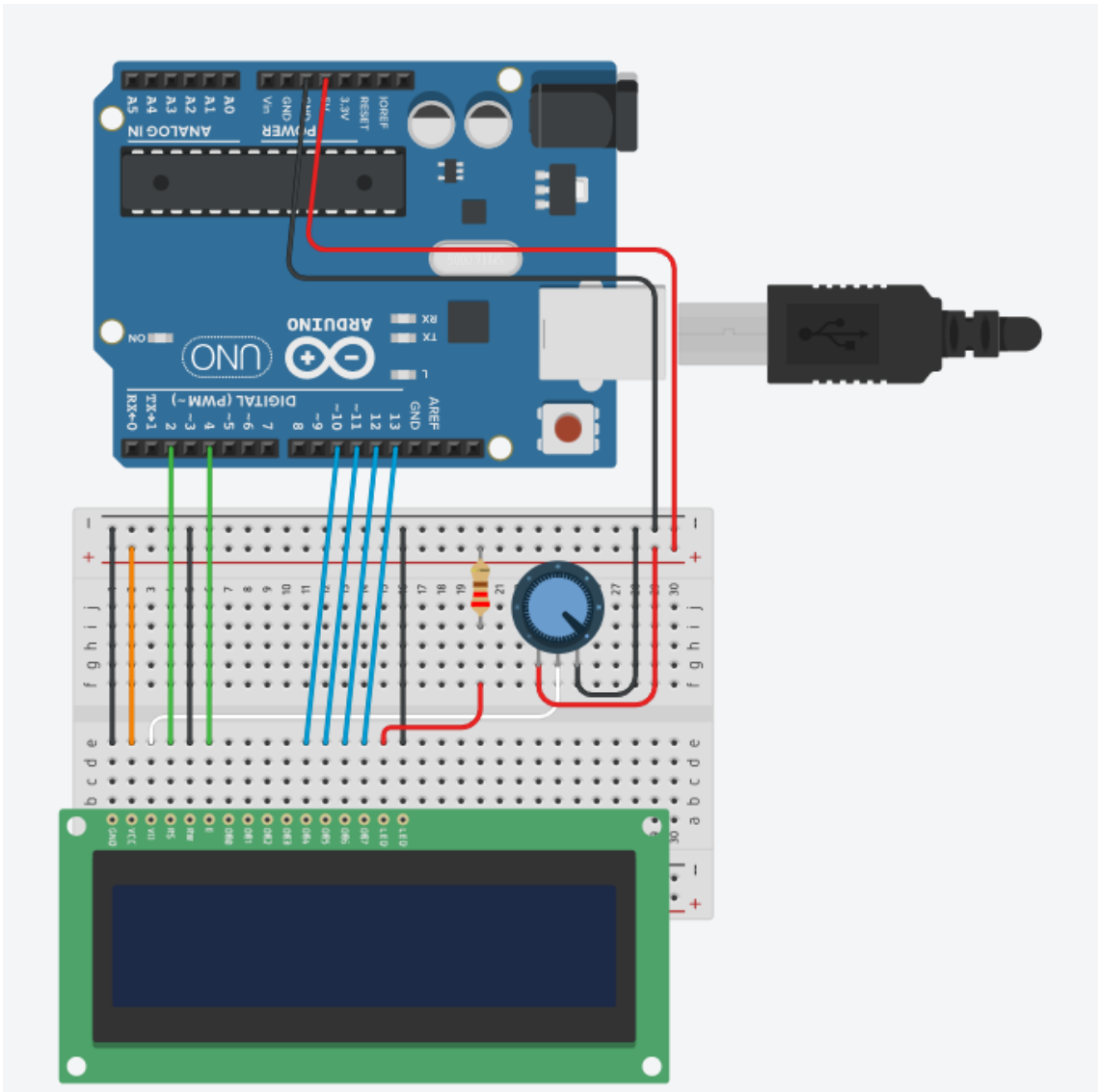
```
  lcd.write('M'); //Escreve apenas um caractere
```

```
}
```

```
void loop()
```

```
{
```

```
}
```



BOTÃO

```
// Ligando LED com push button

// Autor: Luan Alves - www.guiarobotica.com

int pushbutton = 2; // declara o push button na porta 2

int led = 13; // declara led na porta 13

bool estadoled = 0; // variavel de controle


void setup()

{

    pinMode(pushbutton, INPUT_PULLUP); // define o pino do botao como entrada

    pinMode(led, OUTPUT); // define LED como saida

}


void loop()

{

    if (digitalRead(pushbutton) == LOW) // Se o botão for pressionado

    {

        estadoled = !estadoled; // troca o estado do LED

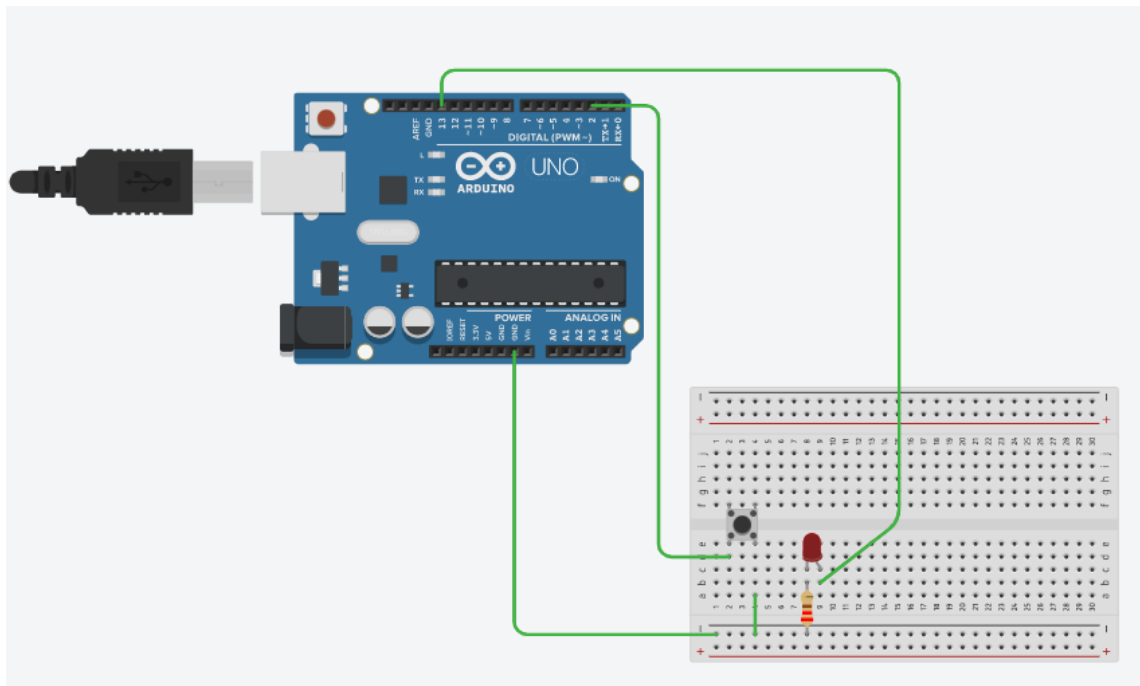
        digitalWrite(led, estadoled);

        while (digitalRead(pushbutton) == LOW);

        delay(100);

    }

}
```



SERVO MOTOR

```
// inclui biblioteca do servomotor

#include <Servo.h>


// define pinos dos servos

#define pinServ1 2

#define pinServ2 3

#define pinServ3 4

#define pinServ4 5


// define as portas dos potenciometros

#define pot1 A0

#define pot2 A1

#define pot3 A2

#define pot4 A3


// nomeia os servos

Servo serv1,serv2,serv3,serv4;


// cria as variáveis dos ângulos de cada motor

int motor1,motor2,motor3,motor4;


unsigned long mostradorTimer = 1;

const unsigned long intervaloMostrador = 5000;


void setup() {
```

```

//inicia o monitor serial

Serial.begin(9600);


// atribui pinos dos servos

serv1.attach(pinServ1);

serv2.attach(pinServ2);

serv3.attach(pinServ3);

serv4.attach(pinServ4);

}


void loop(){


// leitura dos potenciometros

motor1 = map(analogRead(pot1),0,1023,0,180);

motor2 = map(analogRead(pot2),0,1023,0,180);

motor3 = map(analogRead(pot3),0,1023,85,180);

motor4 = map(analogRead(pot4),0,1023,0,43);


// posicionamento dos potenciometros

serv1.write(motor1);

serv2.write(motor2);

serv3.write(motor3);

serv4.write(motor4);


if ((millis() - mostradorTimer) >= intervaloMostrador) {

```

```

// envio para o monitor serial do posicionamentos dos motores

Serial.println("*****");

Serial.print("Pot1:");
Serial.print(analogRead(pot1));
Serial.print(" Angulo Motor1:");
Serial.println(motor1);


Serial.print("Pot2:");
Serial.print(analogRead(pot2));
Serial.print(" Angulo Motor2:");
Serial.println(motor2);


Serial.print("Pot3:");
Serial.print(analogRead(pot3));
Serial.print(" Angulo Motor3:");
Serial.println(motor3);


Serial.print("Pot4:");
Serial.print(analogRead(pot4));
Serial.print(" Angulo Motor4:");
Serial.println(motor4);

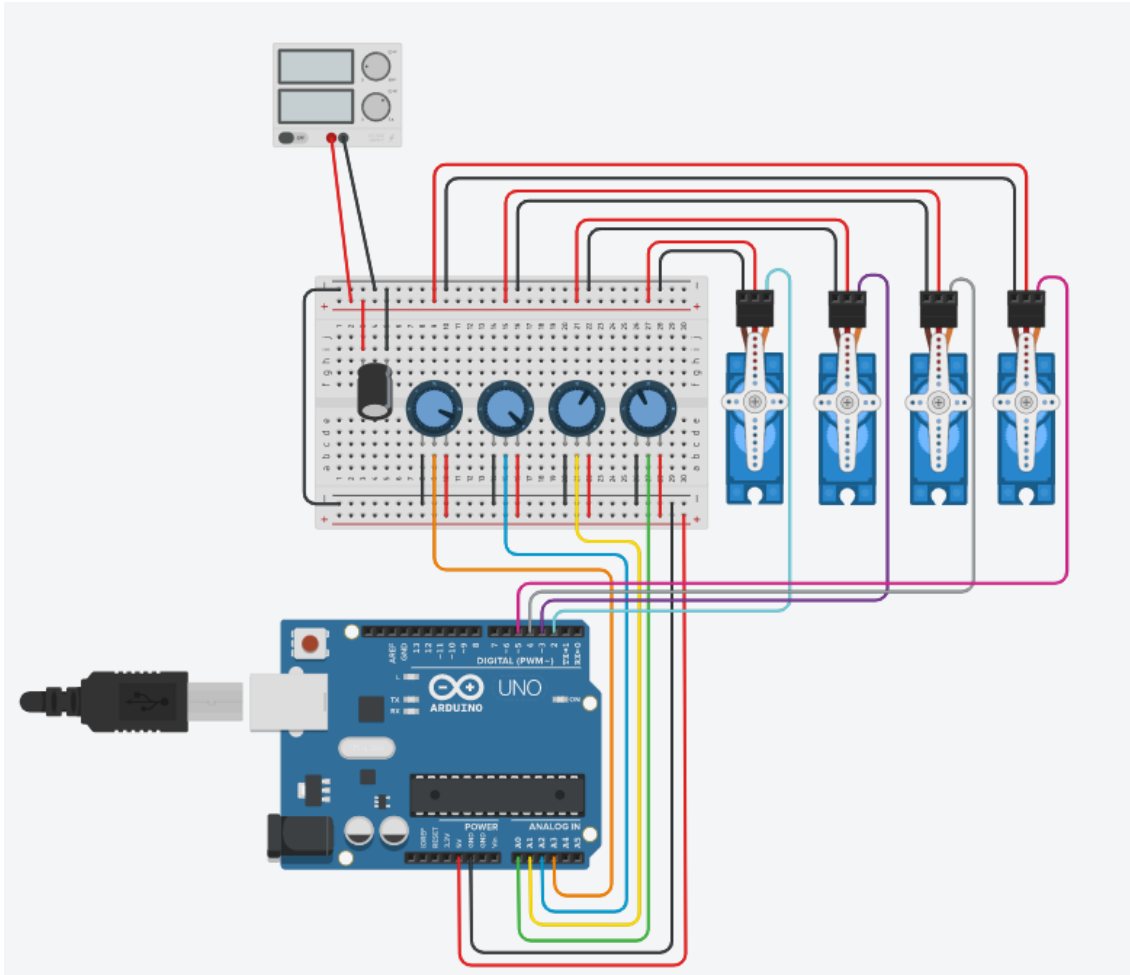

mostradorTimer = millis();
}

// tempo de espera para recomeçar

```

```
delay(100);
```

```
}
```



SERVO MOTOR COM POTENCIÔMETRO

```
#include <Servo.h>
```

```
int pos;
```

```
Servo servomotor;
```

```
int angulo;
```

```
void setup()
```

```
{
```

```
servomotor.attach(7);          //Saída de dados para o servo motor - Porta7
```

```
Serial.begin(9600);
```

```
pos = 0;
```

```
}
```

```
void loop()
```

```
{
```

```
int pot = analogRead(0);          //Faz a leitura do potenciômetro
```

```
angulo = map(pot, 0, 1023, 0, 180); //Relaciona a leitura do potenciômetro
```

```
                                     //com 180° de movim. do servo motor
```

```
//servomotor.write(angulo);
```

```
for (pos = angulo; pos <= 180; pos += 1) {
```

```
servomotor.write(pos);
```

```
delay(15);
```

```
}
```

```

for (pos = angulo; pos >= 0; pos -= 1) {

  servomotor.write(pos);

  delay(30);

}

```

```

Serial.print("Leitura: ");

```

```

//Escreve no Monitor Serial

```

```

Serial.println(angulo);

```

```

//Mostra as leituras de Ângulos no Monitor Serial

```

```

delay(30);

```

```

//Aguarda 30ms até o servo motor chegar na

```

```

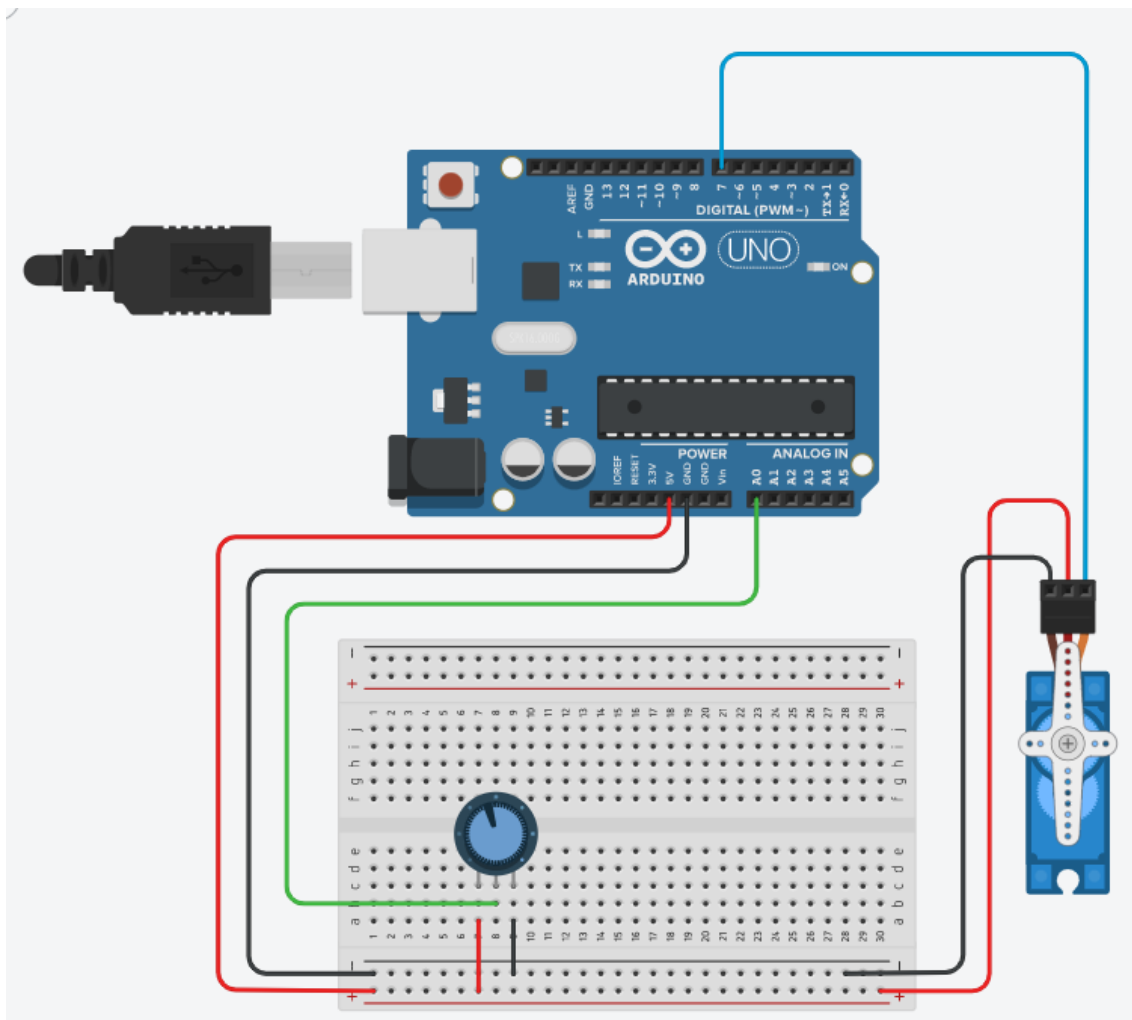
posição

```

```

}

```



MOTOR DC

```
// Declare as variáveis de leitura dos componentes

const int botao = 12;

const int motor = 8;

const int led = 4;


// Crie uma variável para armazenar o estado do switch

int estadobotao = 0;


void setup() {

    // Aqui, declare os componentes como entrada/saída

    pinMode(botao, INPUT_PULLUP);

    pinMode(motor, OUTPUT);

    pinMode(led, OUTPUT);


    //Inicialize a comunicação com o monitor serial

    Serial.begin(9600);

}


void loop() {

    // Faz a leitura do estado do botão e armazena na respectiva variável

    estadobotao = digitalRead(botao);


    // Se o botão estiver pressionado

    if(estadobotao == HIGH){
```

```
digitalWrite(led, HIGH);

digitalWrite(motor, HIGH); // Aciona o motor

// E aciona o LED

Serial.println("Estado atual: motor ligado!");

}


// Caso isso não aconteça

else{

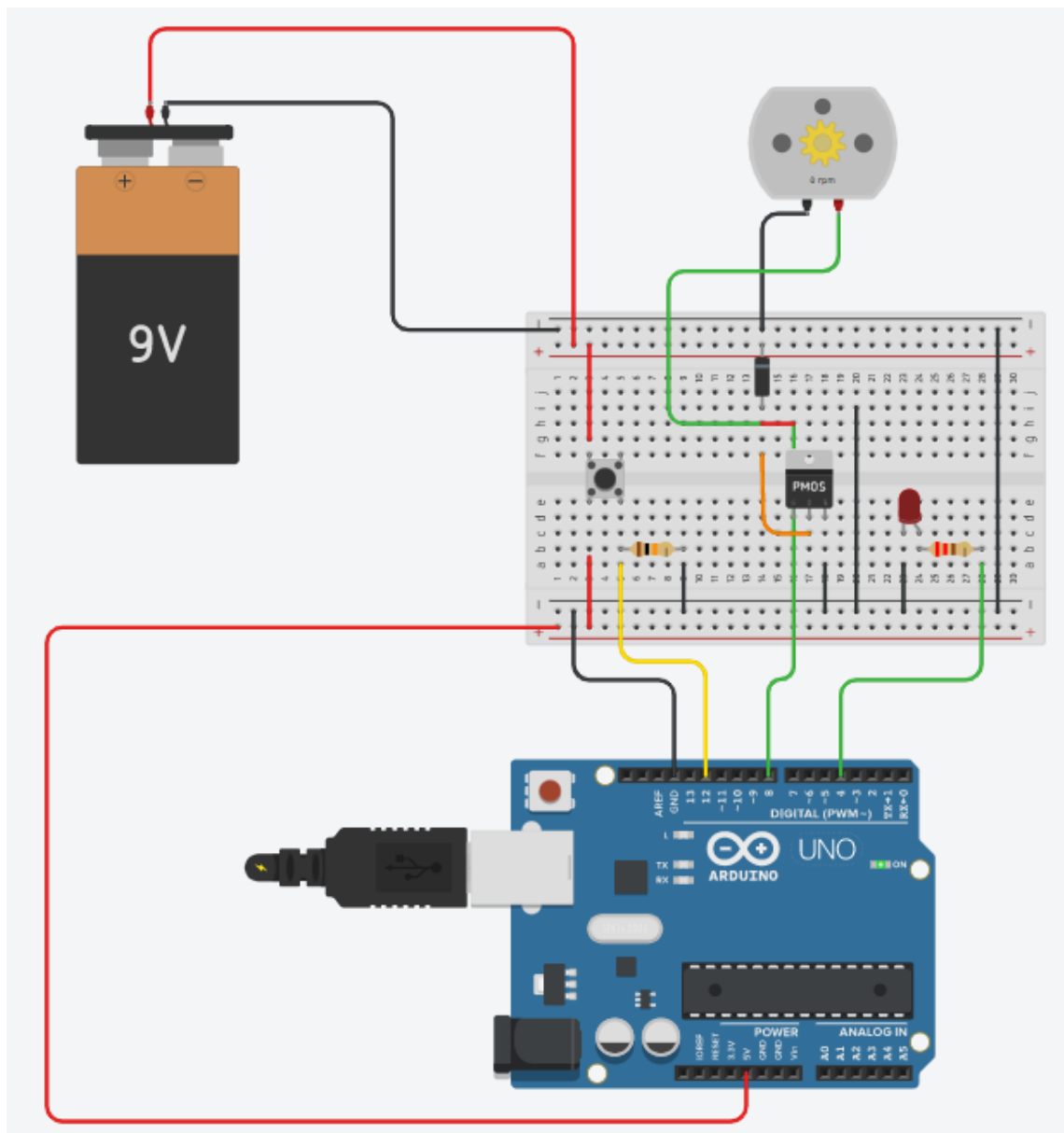
    digitalWrite(motor, LOW); // Motor fica desligado

    digitalWrite(led, LOW); // E o LED também

    Serial.println("Estado atual: motor desligado!");

}

}
```



SENSOR DE LUZ

```
// C++ code

//

/*
  This will make your servo and LED move when
  light is detected.
*/

#include <Servo.h>

int light = 0;

Servo servo_9;

void setup()
{
  pinMode(A0, INPUT);
  servo_9.attach(9, 500, 2500);
  pinMode(12, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  // This sets a variable "light" to the reading from
  // the sensor
  light = analogRead(A0);
  if (light > 500) {
```

```

// while light is detected this loop will run

// First here servo will go to 0 deg. and led is on
servo_9.write(0);
digitalWrite(12, HIGH);

// These print lines are here to allow you to see
// the sensor reading in the serial monitor
Serial.println(light);

delay(500); // Wait for 500 millisecond(s)

// after waiting for a second the servo will go to
// 90 deg and led will go off
servo_9.write(90);
digitalWrite(12, LOW);

// These print lines are here to allow you to see
// the sensor reading in the serial monitor
Serial.println(light);

delay(500); // Wait for 500 millisecond(s)
} else {

// If there is low light this will run and print
// the sensor value to serial monitor (no action on
// servo or led)
Serial.println(light);

delay(500); // Wait for 500 millisecond(s)
}
}

```


SENSOR DE DISTÂNCIA

```
int trig = 13;
```

```
int echo = 12;
```

```
float distancia;
```

```
void setup()
```

```
{
```

```
  pinMode (trig, OUTPUT);
```

```
  pinMode (echo, INPUT);
```

```
  Serial.begin (9600);
```

```
}
```

```
void loop()
```

```
{
```

```
  digitalWrite(trig, LOW);
```

```
  delay(0005);
```

```
  digitalWrite(trig, HIGH);
```

```
  delay(0010);
```

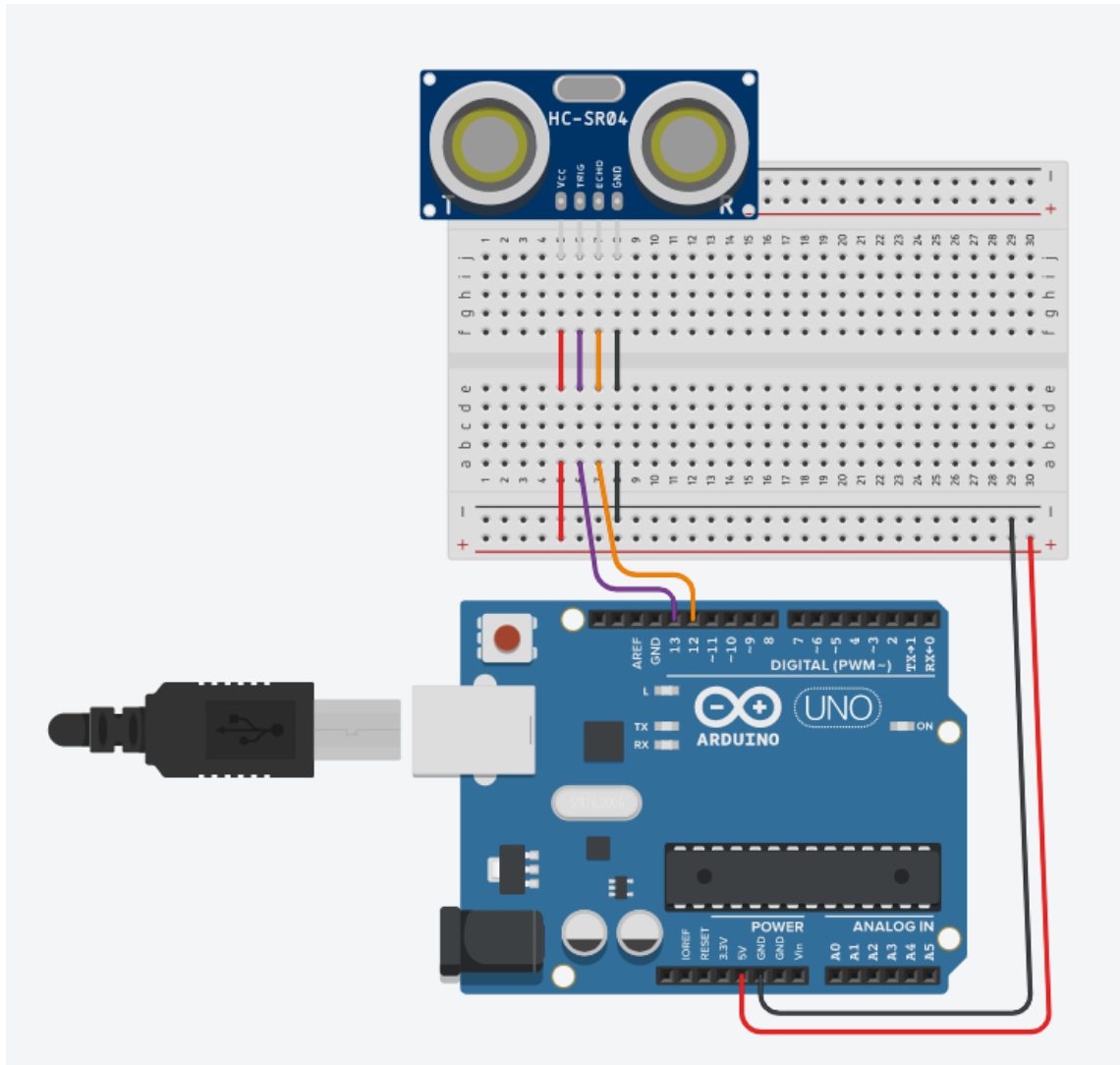
```
  digitalWrite(trig, LOW);
```

```
  distancia = pulseIn (echo, HIGH);
```

```
  distancia = distancia/58;
```

```
  Serial.println (distancia);
```

}



PIEZO

```
int buzzer = 8;

void setup()
{
    // Defines the Buzzer pin as output
    pinMode(buzzer,OUTPUT);
}

void loop()
{

    tone(buzzer,261);

    delay(200);
    //Turns the buzzer off
    noTone(buzzer);

    tone(buzzer,293);
    delay(200);
    noTone(buzzer);

    tone(buzzer,329);
    delay(200);
    noTone(buzzer);

    tone(buzzer,349);
    delay(200);

    tone(buzzer,201);

    delay(200);
    //Turns the buzzer off
```

}



BUZINA

```
const byte buzzer = 2;

void setup()
{
  pinMode(buzzer, OUTPUT);
}

void loop()
{
  tone(buzzer, 440);

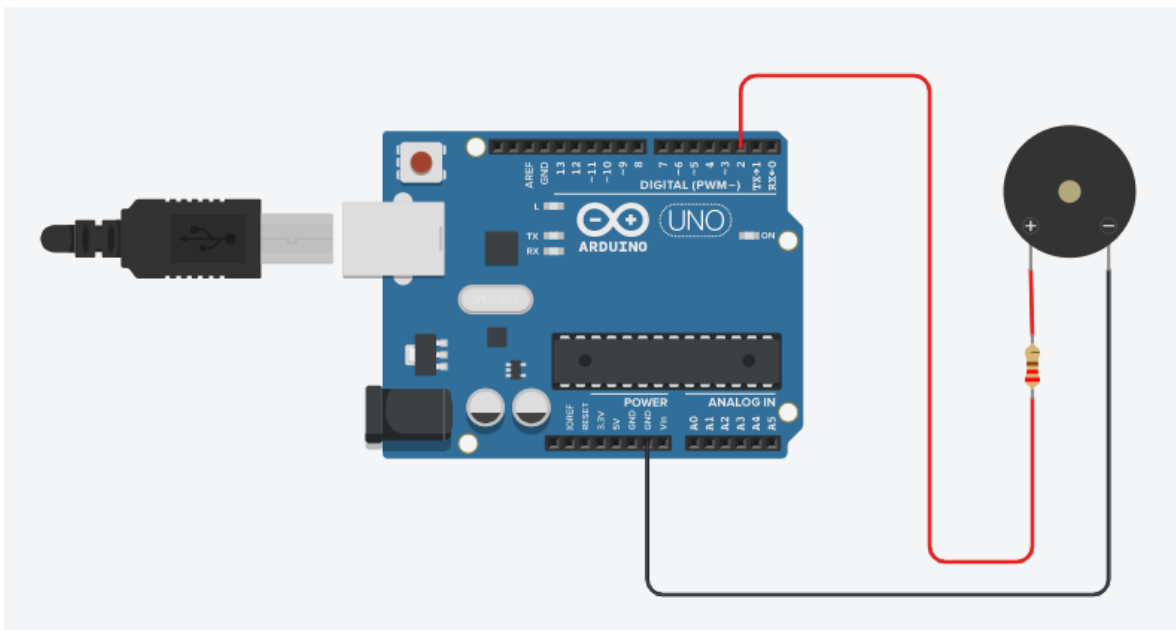
  digitalWrite(LED_BUILTIN, LOW);

  delay(200);

  noTone(buzzer);

  tone(buzzer, 440,100);

  delay(200);
}
```



TECLADO MUSICAL

```
int but1 = 2;
```

```
int but2 = 3;
```

```
int but3 = 4;
```

```
int but4 = 5;
```

```
int but5 = 6;
```

```
int but6 = 7;
```

```
int but7 = 8;
```

```
int but8 = 9;
```

```
int buzzer = 13;
```

```
void setup()
```

```
{
```

```
  //let's declare the button pins as input
```

```
  pinMode(but1,INPUT);
```

```
  pinMode(but2,INPUT);
```

```
  pinMode(but3,INPUT);
```

```
  pinMode(but4,INPUT);
```

```
  pinMode(but5,INPUT);
```

```
  pinMode(but6,INPUT);
```

```
  pinMode(but7,INPUT);
```

```
  pinMode(but8,INPUT);
```

```
  //declare buzzer pin as output
```

```
  pinMode(buzzer,OUTPUT);
```

```
}
```

```

void loop()
{
    // read the value from buttons

    int b1 = digitalRead(but1);

    int b2 = digitalRead(but2);

    int b3 = digitalRead(but3);

    int b4 = digitalRead(but4);

    int b5 = digitalRead(but5);

    int b6 = digitalRead(but6);

    int b7 = digitalRead(but7);

    int b8 = digitalRead(but8);


    if( b1 == 1 ){

        tone(buzzer,300,100);

    }

    if( b2 == 1 ){

        tone(buzzer,400,100);

    }

    if( b3 == 1 ){

        tone(buzzer,500,100);

    }

    if( b4 == 1 ){

        tone(buzzer,600,100);

    }

    if( b5 == 1 ){

        tone(buzzer,700,100);

    }
}

```

```

if( b6 == 1 ){
    tone(buzzer,800,100);
}

if( b7 == 1 ){
    tone(buzzer,900,100);
}

if( b8 == 1 ){
    tone(buzzer,1000,100);

// now let's put a short delay for a nice pitch
    delay(10);

}

```

